GroupProject

2025-11-15

We start by loading the College Scorecard data, restricting to 4-year institutions with non-missing retention and at least 100 undergrads, and constructing our treatment: an indicator for "high-aid" schools, defined as having low-income net price (for $0–30k income) below the median.

```r
# Load raw data
df <- read.csv("./College_Scorecard_Raw_Data_05192025/MERGED2023_24_PP.csv")

# 1. Keep 4-year institutions, non-missing retention, and UGDS >= 100
df_clean <- df[
  df$ICLEVEL == 1 &
    !is.na(df$RET_FT4) &
    df$UGDS >= 100,
]

# 2. Construct low-income net price using public vs private columns
df_clean$low_income_np <- ifelse(
  df_clean$CONTROL == 1,
  df_clean$NPT41_PUB,
  df_clean$NPT41_PRIV
)

# 3. Drop rows with missing low-income net price
df_clean <- df_clean[!is.na(df_clean$low_income_np), ]

# 4. Treatment: high aid = low net price (bottom half of low_income_np)
median_np <- median(df_clean$low_income_np, na.rm = TRUE)
df_clean$high_aid <- as.integer(df_clean$low_income_np <= median_np)

# 5. Keep complete cases only for a *core* set of variables we know exist
vars_core <- c("RET_FT4", "high_aid", "CONTROL", "ADM_RATE", "SAT_AVG", "UGDS")
df_clean <- df_clean[complete.cases(df_clean[, vars_core]), ]

# sanity checks
table(df_clean$high_aid)
```

```
##
##   0   1
## 490 556
```

```r
head(df_clean[, c("RET_FT4", "low_income_np", "high_aid")])
```

```
##   RET_FT4 low_income_np high_aid
## 1  0.6387         13776        1
## 2  0.8195         14816        0
## 4  0.8050         16521        0
## 5  0.6045         13512        1
```

```
## 6  0.8609         18592        0
## 9  0.6580         12999        1
```

```r
nrow(df_clean)
```

```
## [1] 1046
```

Next, we define a candidate set of control variables capturing sector, degree type, geography, student composition, and resources. We then restrict this list to variables that (a) actually exist in our cleaned data and (b) are not entirely missing or NA. This list will be used both in OLS and in the causal forest.

```r
# Choose a rich set of candidate controls
covariate_candidates <- c(
  "CONTROL",          # institution control type (public/private nonprofit/for-profit)
  "PREDDEG",          # highest degree primarily offered
  "REGION",           # geographic region of the U.S.
  "LOCALE",           # campus setting (city / suburb / town / rural)
  "ADM_RATE",         # admission rate (share of applicants admitted)
  "SAT_AVG",          # average SAT score of enrolled students
  "UGDS",             # total undergrad enrollment
  "UGDS_BLACK",       # share of undergrads who are Black
  "UGDS_HISP",        # share of undergrads who are Hispanic/Latino
  "UGDS_ASIAN",       # share of undergrads who are Asian
  "UGDS_MEN",         # share of undergrads who are male
  "UG25ABV",          # share of undergrads age 25 or older
  "PPTUG_EF",         # share of undergrads attending part-time
  "PAR_ED_PCT_1STGEN",# share of students who are first-gen (no parent w/ college degree)
  "PCTPELL",          # share of students receiving Pell Grants
  "PCTFLOAN",         # share of students receiving federal loans
  "COSTT4_A",         # average annual total cost of attendance
  "INEXPFTE",         # instructional expenses per FTE student
  "PFTFAC",           # share of faculty who are full-time
  "AVGFACSAL"         # average faculty salary
)

# Keep only variables that (1) exist in df_clean and (2) are not all NA
covariate_candidates <- covariate_candidates[
  covariate_candidates %in% names(df_clean)
]

nonempty_controls <- covariate_candidates[
  colSums(!is.na(df_clean[, covariate_candidates, drop = FALSE])) > 0
]

nonempty_controls
```

```
##  [1] "CONTROL"    "PREDDEG"    "REGION"     "LOCALE"     "ADM_RATE"
##  [6] "SAT_AVG"    "UGDS"       "UGDS_BLACK" "UGDS_HISP"  "UGDS_ASIAN"
## [11] "UGDS_MEN"   "UG25ABV"    "PPTUG_EF"   "PCTPELL"    "PCTFLOAN"
## [16] "COSTT4_A"   "INEXPFTE"   "PFTFAC"     "AVGFACSAL"
```

```r
length(nonempty_controls)
```

```
## [1] 19
```

We then estimate a baseline OLS regression of retention on the high-aid treatment and this set of controls. This gives us a linear benchmark before we move to the more flexible causal forest.

```r
# Build formula string dynamically: RET_FT4 ~ high_aid + controls...
ols_formula_str <- paste(
  "RET_FT4 ~ high_aid +",
  paste(nonempty_controls, collapse = " + ")
)

ols_formula <- as.formula(ols_formula_str)
ols_formula   # just to see what the final formula is
```

```
## RET_FT4 ~ high_aid + CONTROL + PREDDEG + REGION + LOCALE + ADM_RATE +
##     SAT_AVG + UGDS + UGDS_BLACK + UGDS_HISP + UGDS_ASIAN + UGDS_MEN +
##     UG25ABV + PPTUG_EF + PCTPELL + PCTFLOAN + COSTT4_A + INEXPFTE +
##     PFTFAC + AVGFACSAL
```

```r
# Run OLS
ols <- lm(ols_formula, data = df_clean)

summary(ols)
```

```
##
## Call:
## lm(formula = ols_formula, data = df_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.57030 -0.03196  0.00064  0.03490  0.33355
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.299e-01  6.627e-02   4.978 7.54e-07 ***
## high_aid     8.163e-03  5.459e-03   1.495  0.13519
## CONTROL      2.880e-03  8.809e-03   0.327  0.74376
## PREDDEG      3.415e-02  1.439e-02   2.372  0.01787 *
## REGION      -2.645e-03  1.244e-03  -2.127  0.03366 *
## LOCALE       8.102e-05  2.395e-04   0.338  0.73518
## ADM_RATE    -1.396e-02  1.302e-02  -1.072  0.28395
## SAT_AVG      3.289e-04  3.281e-05  10.027  < 2e-16 ***
## UGDS         2.213e-06  3.363e-07   6.581 7.47e-11 ***
## UGDS_BLACK   5.044e-02  1.962e-02   2.571  0.01027 *
## UGDS_HISP    8.713e-02  2.257e-02   3.861  0.00012 ***
## UGDS_ASIAN   2.578e-03  4.597e-02   0.056  0.95528
## UGDS_MEN    -4.140e-02  1.745e-02  -2.373  0.01785 *
## UG25ABV     -6.732e-03  3.367e-02  -0.200  0.84156
## PPTUG_EF    -1.472e-01  3.734e-02  -3.942 8.62e-05 ***
## PCTPELL     -1.939e-01  3.283e-02  -5.907 4.75e-09 ***
## PCTFLOAN    -3.841e-02  1.723e-02  -2.229  0.02604 *
## COSTT4_A     1.410e-07  2.860e-07   0.493  0.62207
## INEXPFTE    -1.665e-07  2.817e-07  -0.591  0.55447
## PFTFAC       1.840e-02  1.013e-02   1.816  0.06970 .
## AVGFACSAL    3.164e-06  1.504e-06   2.104  0.03563 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.0654 on 1019 degrees of freedom
##   (6 observations deleted due to missingness)
```

```
## Multiple R-squared:  0.6496, Adjusted R-squared:  0.6428
## F-statistic: 94.47 on 20 and 1019 DF,  p-value: < 2.2e-16
```

```r
nrow(df_clean)
```

```
## [1] 1046
```

```r
table(df_clean$high_aid)
```

```
##
##   0   1
## 490 556
```

Now we fit a causal forest using the same outcome (RET_FT4), treatment (high_aid), and control set. This model estimates the average treatment effect (ATE), the effect for treated schools (ATT), and an institution-level treatment effect (CATE) for each school.

```r
library(grf)

# Outcome and treatment
Y <- df_clean$RET_FT4
W <- df_clean$high_aid

# Covariate matrix X using the same nonempty controls
X <- as.matrix(df_clean[, nonempty_controls, drop = FALSE])

# Drop any rows with NAs in Y, W, or X
keep_grf <- complete.cases(cbind(Y, W, X))

Y_grf <- Y[keep_grf]
W_grf <- W[keep_grf]
X_grf <- X[keep_grf, , drop = FALSE]

length(Y_grf)   # effective sample size for GRF
```

```
## [1] 1040
```

```r
set.seed(0)
cf <- causal_forest(X_grf, Y_grf, W_grf)

# Overall ATE
average_treatment_effect(cf)
```

```
## Warning in average_treatment_effect(cf): Estimated treatment propensities go as
## high as 0.968 which means that treatment effects for some treated units may not
## be well identified. In this case, using `target.sample=control` may be helpful.
```

```
##    estimate     std.err
## 0.018634432 0.004523826
```

```r
# ATT (treated institutions only)
average_treatment_effect(cf, target.sample = "treated")
```

```
## Warning in average_treatment_effect(cf, target.sample = "treated"): Estimated
## treatment propensities go as high as 0.968 which means that treatment effects
## for some treated units may not be well identified. In this case, using
## `target.sample=control` may be helpful.
```

```
##    estimate     std.err
```

```
## 0.018507650 0.006558597
```

```r
# Institution-level CATEs
cate_hat <- predict(cf)$predictions
summary(cate_hat)
```

```
##     Min.  1st Qu.   Median     Mean  3rd Qu.     Max.
## 0.002765 0.014315 0.016682 0.016967 0.019196 0.034789
```

Because causal forests rely on overlap in the estimated propensity scores, we inspect the distribution of the estimated treatment propensities and perform an initial trim (keeping schools with propensities between 0.05 and 0.95). We then re-estimate the causal forest and recompute ATE and ATT on this trimmed sample.

```r
# Estimated propensity scores
p_hat <- cf$W.hat

summary(p_hat)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.1479  0.2310  0.4377  0.5323  0.8718  0.9685
```

```r
quantile(p_hat, c(0.01, 0.05, 0.95, 0.99))
```

```
##        1%        5%       95%       99%
## 0.1613316 0.1740389 0.9329207 0.9544074
```

```r
# Keep only institutions with reasonably good overlap
keep_trim <- p_hat >= 0.05 & p_hat <= 0.95
sum(keep_trim)
```

```
## [1] 1024
```

```r
Y_trim <- Y_grf[keep_trim]
W_trim <- W_grf[keep_trim]
X_trim <- X_grf[keep_trim, , drop = FALSE]

cf_trim <- causal_forest(X_trim, Y_trim, W_trim)

tau_all     <- average_treatment_effect(cf_trim)
```

```
## Warning in average_treatment_effect(cf_trim): Estimated treatment propensities
## go as high as 0.968 which means that treatment effects for some treated units
## may not be well identified. In this case, using `target.sample=control` may be
## helpful.
```

```r
tau_treated <- average_treatment_effect(cf_trim, target.sample = "treated")
```

```
## Warning in average_treatment_effect(cf_trim, target.sample = "treated"):
## Estimated treatment propensities go as high as 0.968 which means that treatment
## effects for some treated units may not be well identified. In this case, using
## `target.sample=control` may be helpful.
```

```r
tau_all
```

```
##    estimate     std.err
## 0.017570715 0.004540491
```

```r
tau_treated
```

```
##    estimate     std.err
## 0.016121391 0.006790703
```

As a robustness check, we apply more aggressive trimming (keeping only schools with propensities between 0.10 and 0.90), re-fit the causal forest, and examine how the ATE and ATT change. This focuses estimation on institutions with the strongest overlap.

```r
# more aggressive trimming
keep_trim2 <- p_hat >= 0.10 & p_hat <= 0.90
sum(keep_trim2)
```

```
## [1] 871
```

```r
Y_trim2 <- Y_grf[keep_trim2]
W_trim2 <- W_grf[keep_trim2]
X_trim2 <- X_grf[keep_trim2, , drop = FALSE]

cf_trim2 <- causal_forest(X_trim2, Y_trim2, W_trim2)

average_treatment_effect(cf_trim2)
```

```
##    estimate     std.err
## 0.019320922 0.004881964
```

```r
average_treatment_effect(cf_trim2, target.sample = "treated")
```

```
##    estimate     std.err
## 0.021367470 0.007433431
```

```r
summary(cf_trim2$W.hat)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.1519  0.2151  0.3084  0.4544  0.7849  0.9177
```

```r
quantile(cf_trim2$W.hat, c(0.01, 0.05, 0.95, 0.99))
```

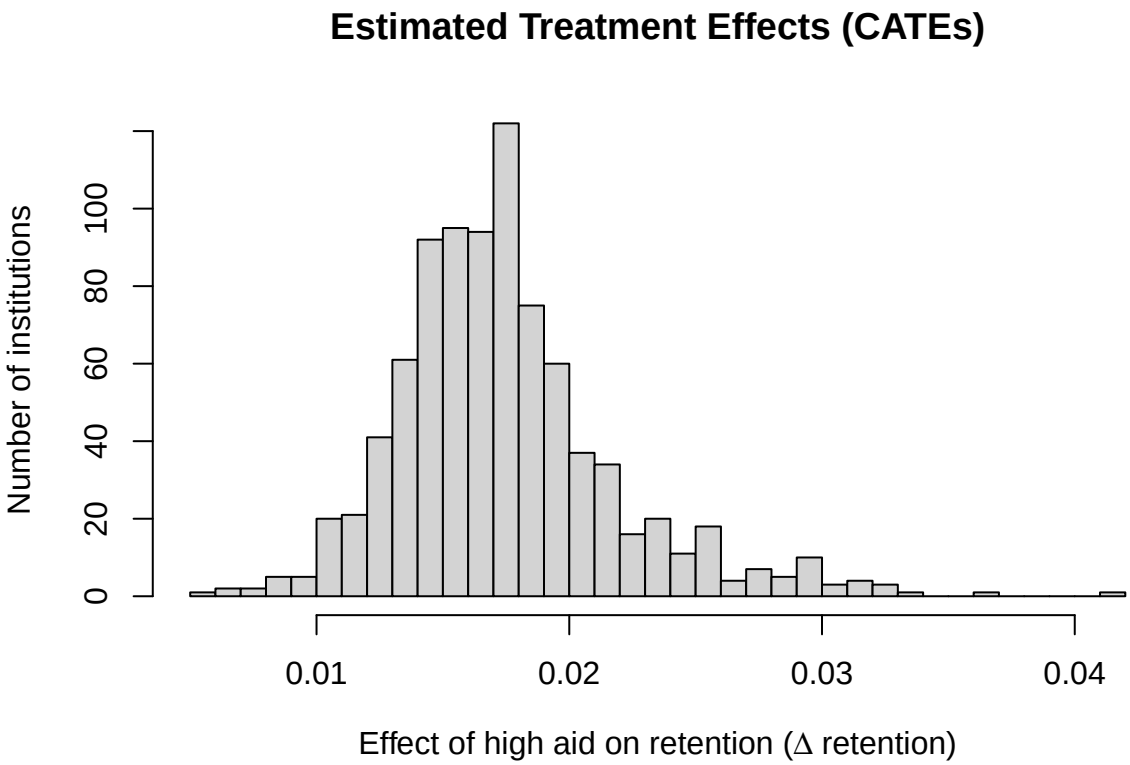
```
##        1%        5%       95%       99%
## 0.1577971 0.1692604 0.8787433 0.9045822
```

To interpret the forest, we link the trimmed forest back to the rows of df_clean, so that each institution's estimated treatment effect (CATE) can be matched with its observed characteristics. This prepares the data for plots of the distribution and heterogeneity of the estimated effects.

```r
# Indices linking back to df_clean
idx_grf   <- which(keep_grf)      # rows of df_clean used in the first forest
idx_trim2 <- idx_grf[keep_trim2]  # rows of df_clean used in cf_trim2

df_trim2 <- df_clean[idx_trim2, ]

# Estimated CATEs for the trimmed sample
cate_trim2 <- predict(cf_trim2)$predictions
```

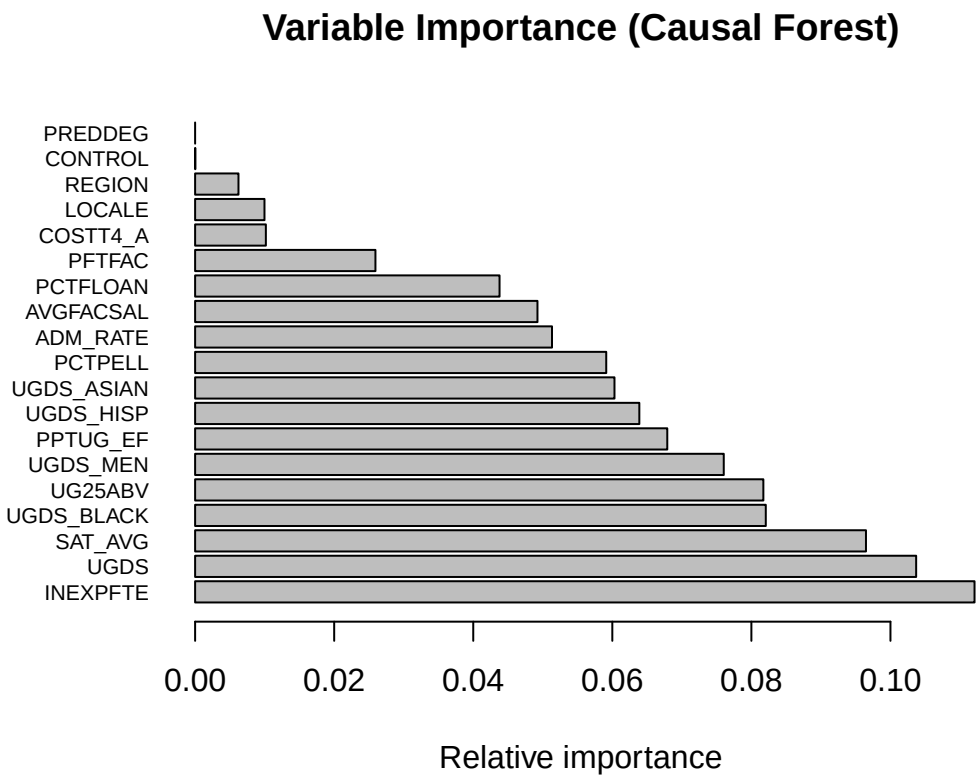
We first plot a histogram of the estimated CATEs to see the distribution of treatment effects across institutions (how much high aid seems to help retention at different schools).

```r
hist(
  cate_trim2,
  breaks = 30,
  main   = "Estimated Treatment Effects (CATEs)",
  xlab = expression("Effect of high aid on retention ("*Delta*" retention)"),
  ylab   = "Number of institutions"
)
```



Next, we plot variable importance from the causal forest, which summarizes which covariates are most useful for predicting heterogeneity in the treatment effect (which characteristics the forest "cares about" when splitting).

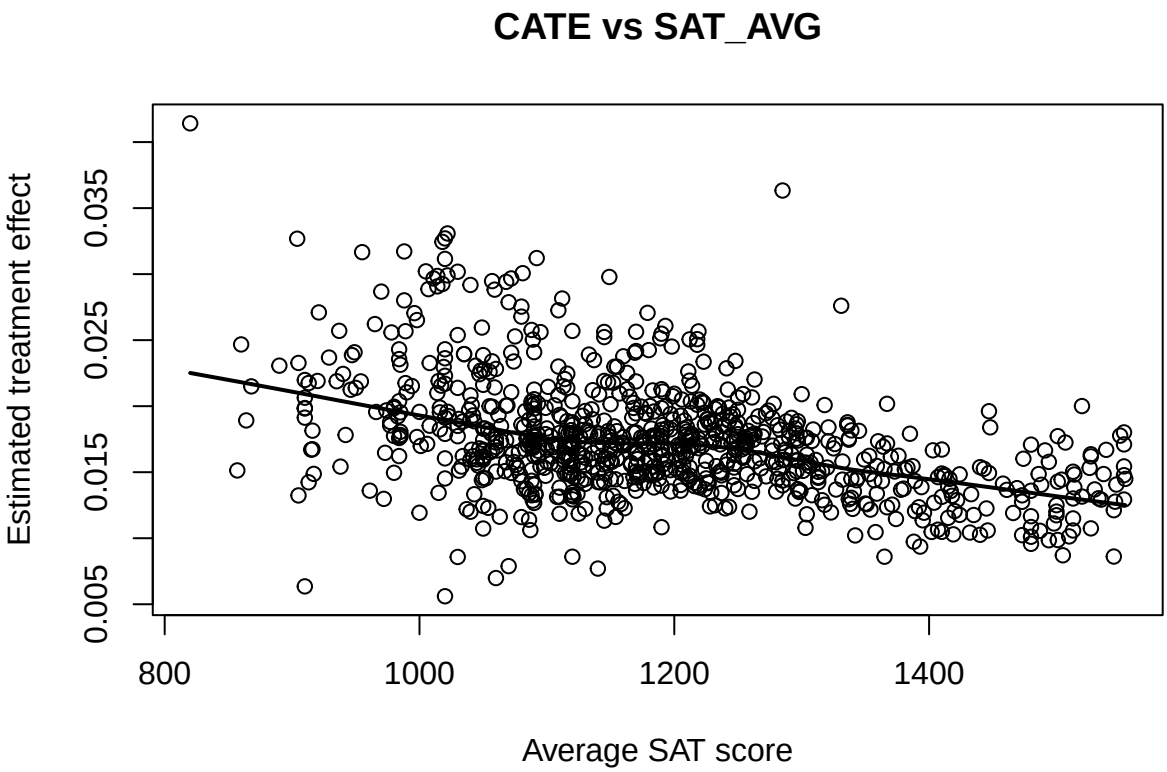
```
# Raw importance from the causal forest
vi <- variable_importance(cf_trim2)
names(vi) <- nonempty_controls  # label with covariate names

# Sort variables by importance (largest at top)
vi_sorted <- sort(vi, decreasing = TRUE)

# Make room on the left for labels
old_par <- par(mar = c(5, 10, 4, 2))  # bottom, left, top, right

barplot(
  vi_sorted,
  horiz      = TRUE,             # horizontal bars
  names.arg  = names(vi_sorted), # show variable names
  las        = 1,                # labels horizontal
  cex.names  = 0.7,
  main       = "Variable Importance (Causal Forest)",
  xlab       = "Relative importance"
)
```




```r
par(old_par)  # reset margins
```

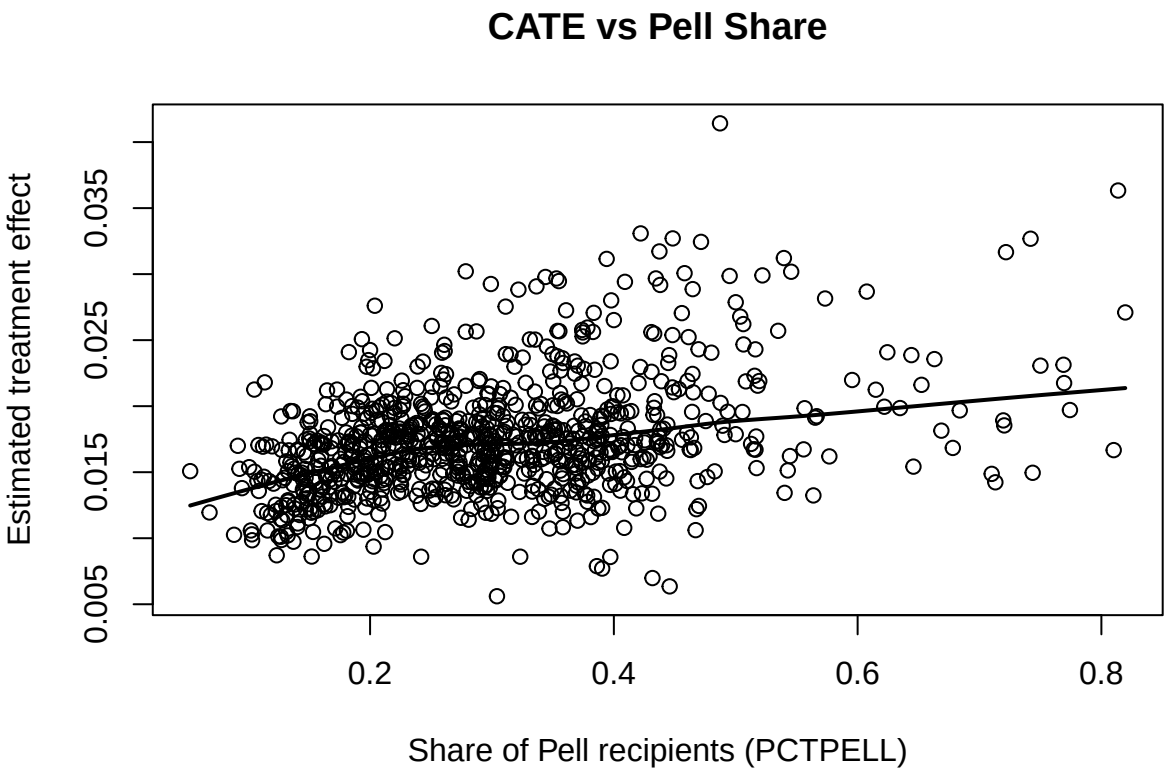
We then plot CATE vs SAT_AVG to examine whether the effect of high aid is larger at institutions with higher or lower average SAT scores—that is, whether aid seems to matter differently by academic preparation.

```r
plot(
  df_trim2$SAT_AVG, cate_trim2,
  xlab = "Average SAT score",
  ylab = "Estimated treatment effect",
  main = "CATE vs SAT_AVG"
)
lines(lowess(df_trim2$SAT_AVG, cate_trim2), lwd = 2)
```



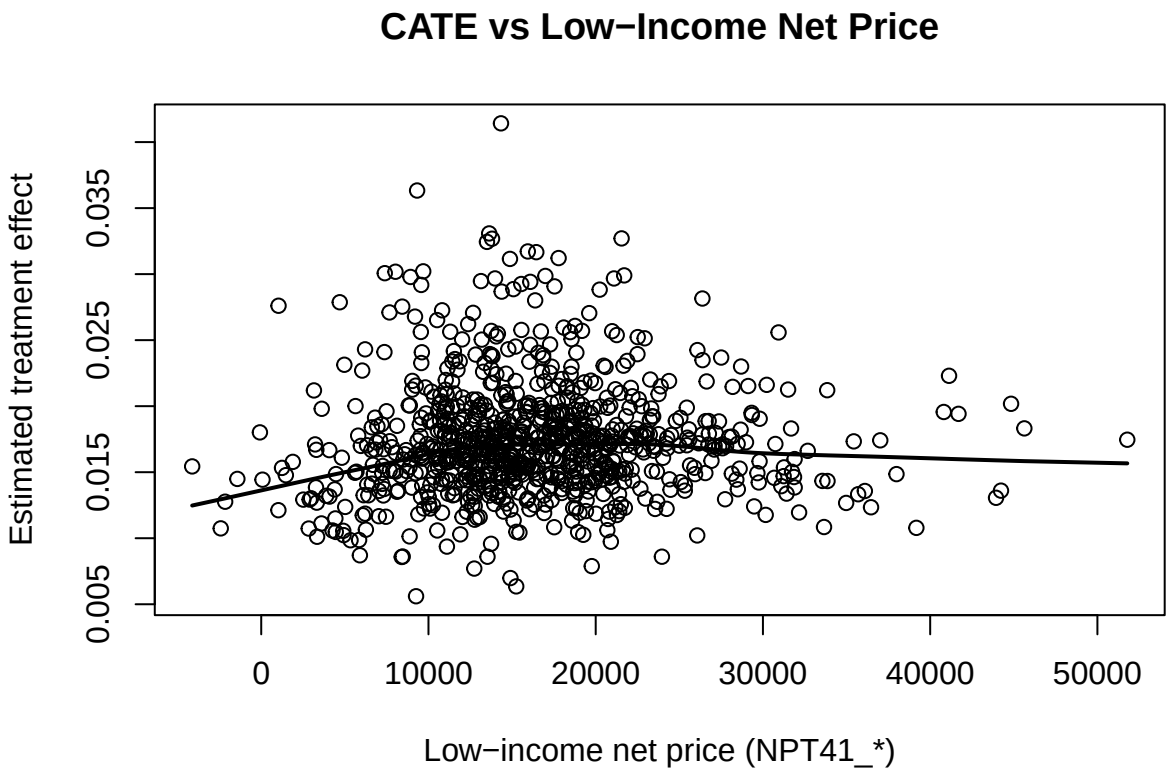
We also look at CATE vs Pell share (PCTPELL) to see whether high-aid policies appear more beneficial at institutions that serve a larger share of Pell recipients.

```
plot(
  df_trim2$PCTPELL, cate_trim2,
  xlab = "Share of Pell recipients (PCTPELL)",
  ylab = "Estimated treatment effect",
  main = "CATE vs Pell Share"
)
lines(lowess(df_trim2$PCTPELL, cate_trim2), lwd = 2)
```



Finally, we plot CATE vs low-income net price, the variable that defines our treatment. This helps visualize whether the estimated gains from being high-aid vary smoothly with the underlying net price level.

```
plot(
  df_trim2$low_income_np, cate_trim2,
  xlab = "Low-income net price (NPT41_*)",
  ylab = "Estimated treatment effect",
  main = "CATE vs Low-Income Net Price"
)
lines(lowess(df_trim2$low_income_np, cate_trim2), lwd = 2)
```


CATE vs Low−Income Net Price
Estimated treatment effect
0.005
0.015
0.025
0.035
0
10000
20000
30000
40000
50000
Low−income net price (NPT41_*)